Cross-Dataset Model Training for Hyperspectral Image Classification Using Self-Supervised Learning

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***Abstract*— With the development of deep learning and the increase in the amount of data, general artificial intelligence models have become a popular research area nowadays. When facing a new application scenario, a pretraining general model can often show better performance than models trained with new data on its own. However, because of the specificity of the differences in hyperspectral image data bands, the current hyperspectral image classification (HSIC) field has not proposed a better general model training solution, and it is difficult to utilize the information of the existing hyperspectral datasets for model training in the face of a new scenario. In order to solve this problem, this article proposes a generalized hyperspectral classification model training method, which effectively completes the training of hyperspectral classification models across datasets by adaptive channel module and masked self-supervised pretraining method, and can pretrain and fine-tune hyperspectral classification models using multiple datasets. The adaptive channel module is able to solve the band difference problem of using hyperspectral datasets across datasets, and the masked self-supervised learning method solves the label difference and labeling difficulties of training models across datasets. Experimental results on multiple datasets show that the method proposed in this article can effectively use a large amount of data to complete the pretraining of hyperspectral classification models, and the fine-tuning results on downstream datasets have certain advantages relative to current advanced deep learning methods.**

Received 14 May 2024; revised 23 July 2024 and 9 October 2024; accepted 29 October 2024. Date of publication 8 November 2024; date of current version 20 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62276206, Grant 62231027, and Grant 62272152; in part by the Aeronautical Science Foundation of China under Grant 2023Z071081001; in part by the Key Research and Development Program of Shaanxi Province under Grant 2022GY-062; in part by the Key Research and Development Program of Hunan Province under Grant 2022GK2020; in part by the Cultivation Project of Yuelu Mountain Industrial Innovation Center under Grant 2023YCII0123; in part by the Natural Science Foundation Project of Chongqing, Chongqing Science and Technology Commission under Grant CSTB2024NSCQ-MSX0920; and in part by the National Key Laboratory of Electromagnetic Space Security. *(Corresponding authors: Jing Bai; Zhu Xiao.)*

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Digital Object Identifier 10.1109/TGRS.2024.3493969

***Index Terms*— Classification, cross dataset, general model, hyperspectral image, masked autoencoder, self-supervised learning.**

I. Introduction

HYPERSPECTRAL image is a type of remote sensing image that contains a continuous spectrum, providing abundant spectral information. Each pixel in a hyperspectral image typically contains hundreds to thousands of spectral bands between visible and infrared regions [1]. Hyperspectral image classification (HSIC) is a pixel-level classification task that involves classifying each individual pixel into different categories or classes via using spectral information from multiple spectral bands. Compared to traditional remote sensing images, hyperspectral images provide more comprehensive spectral information, facilitating HSIC in various application domains such as agriculture, environmental monitoring, and urban planning [2], [3], [4], [5]. On the downside, given the large number of spectral bands and the complex pixel-level processing [6], it is challenging to accomplish an efficient and high-accuracy HSIC task.

After decades of development, deep learning has become a widely used method in HSIC tasks [7], [8], [9], [10], [11], [12]. Facing different datasets, most existing methods use supervised learning to train a model individually to achieve high classification accuracy. Those methods usually use hyperspectral images with pixel-level labels to build training sets and have produced good results [11], [13], [14]. Nevertheless, the supervised learning methods fail to resolve the problem of dependence on annotation data. Such problem becomes severer in the new remote sensing applications, as it is extremely difficult to collect enough annotated samples. To tackle this issue and achieve simpler HSIC, one feasible solution is to train a cross-dataset model for HSIC using unlabeled data and cross datasets without labels.

Although promising, training cross-dataset models for HSIC tasks is not straightforward to accomplish. First, there are inherent issues with applying data across datasets, such as inconsistent labels. In addition, hyperspectral images are usually acquired using hyperspectral imagers, spectral radiometer, or spectrometers. Unlike natural images with three channels, due to the differences in hyperspectral acquisition equipment, different hyperspectral image datasets typically have different combinations of band channels. Supervised learning usually

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has strict requirements on the input and output shapes of neural networks, which makes it difficult for supervised learning methods to be applied to training HSIC models across datasets. Recent works [15], [16], [17] have used domain adaptation to complete the application of an HSIC model across datasets, using information from the source domain dataset as an aid to improve the classification performance of the target dataset. Other works attempt to reduce the dependence on labeled samples by using self-supervised learning methods that can obtain essential features [18], [19], [20], [21]. These methods help to improve the performance of HSIC tasks. However, domain adaptation methods can only be used in the source and target domains rather than multiple datasets. Due to spectral differences between datasets, self-supervised learning methods have not been properly applied to cross-dataset problems. Summarizing, we face the following technical issues for training cross-dataset models in HSIC tasks.

1) Given the difficulty in collection and annotation, it is impractical to collect sufficient labeled samples for HSIC model training in real-world application scenarios. Although much labeled data exists, it is challenging to apply the existing datasets in new scenarios. Due to band differences and labeling differences between hyperspectral image datasets, it is difficult to achieve cross-dataset applications in HSIC tasks. The existing classification models can only use transfer learning and domain adaptation to obtain information from another dataset, while failing to utilize multiple datasets for model training.
2) Hyperspectral image has a more complex structure and feature representation compared to traditional image data, and the existing natural image models usually obtain features from spatial information, while hyperspectral image has more adequate spectral information. Feature extraction networks that have good results in the field of traditional natural images inevitably obtain sufficient information from hyperspectral image. How to combine the spectral features of hyperspectral data itself while considering the spatial features of the image is one of the important issues in the field of HSIC.
3) The currently commonly used masked self-supervised learning uses an approach where the original image is masked in some way, and the encoder and decoder act as an overall network that recovers the original image. And in the overall image recovery process, the encoder undertakes the task of feature extraction and the decoder undertakes the task of recovering the image from features [22]. The conventional design does not differentiate the tasks and limits the feature extraction performance of the encoder.

To address the abovementioned challenges and fully leverage the knowledge from cross dataset, we propose a novel method for training cross-dataset hyperspectral classification model that can efficiently extract spectral features by utilizing multiple datasets. Specifically, we introduce a cross-dataset self-supervised training approach, which adapts to different spectral datasets using adaptive layer convolution and conducts self-supervised training using a masked autoencoder. Self-supervised learning can utilize unlabeled data to complete model training, effectively avoiding label inconsistency issues when training models across datasets. This enables large-scale model training for HSIC task across datasets. This method achieves classification performance surpassing supervised learning with minimal sample fine-tuning. To validate the effectiveness of our method, we conduct comprehensive experimental comparisons on five datasets. The main contributions of this article are as follows.

1) We present a new method for training HSIC models across datasets based on adaptive channel modules and masked autoencoder. Our method utilizes a large amount of unlabeled hyperspectral data to improve the classification performance and feature extraction robustness of the model. Based on the characteristics of hyperspectral image data, the data is segmented in 3-D and self-supervised learning is used to mask and recover the segmented image blocks. This approach can be considered as a new solution for generalized HSIC models.
2) We propose a two-branch Transformer architecture that can better combine spatial–spectral information to improve the shortcomings in the utilization of hyperspectral data information. Compared with the traditional Vision Transformer architecture, this spatial–spectral Transformer, which combines two-branch Transformer and cross-attention mechanism, is able to extract the spatial and spectral information individually and this spatial–spectral Transformer can extract spatial information and spectral information separately, and perform feature fusion according to the correlation between the features, thus achieving more robust feature extraction performance.
3) We propose a context-based masked self-supervision approach. This approach tries to optimize the parameters of the encoder network in the intermediate feature stage to obtain more separable encoder output features, separating the feature extraction task from the image restoration task thus differentiating the encoder training from the decoder training, and being able to obtain better encoder parameters.

The remaining parts of this article are organized as follows. Section II presents related works. Section III describes the proposed classification architecture. Section IV presents and discusses the conducted experiments and studies. Finally, Section V provides the conclusions of this work with an outlook on future research lines.

II. Related Works

A. Hyperspectral Image Classification

Although the task of hyperspectral classification suffers from problems such as lack of sufficient data and complex spectral channels, with the development of the field of machine learning, many effective methods have been proposed in the field of HSIC. Traditional machine learning algorithms such

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as K-nearest neighbor (KNN) methods [23], support vector machines (SVMs) [24], and random forests have been applied to the field of HSIC with some success. With the emergence of deep learning and convolutional neural networks (CNNs), CNNs have been widely used in HSIC tasks due to their localized feature extraction capability and translation invariance, and many brand-new results have emerged in recent years [25], [26]. In some work, 2D-CNN has been used to obtain both image information and spectral information to accomplish the classification task [27], [28]. 3D-CNN has also been used in some HSIC work due to its ability to capture more connections between spectral channels [29], [30]. Some research even combines 2D-CNN and 3D-CNN to accomplish the task of HSIC and achieves relatively good results [31].

At the same time, recurrent neural networks (RNNs) and generative adversarial network (GAN) [32] have also seen certain applications in the field of HSIC. In [33], a three-player spectral GAN architecture was proposed to afford GAN the ability to manage minority classes under imbalanced conditions. Very recently, graph CNNs (GCNs), due to their excellent modeling capabilities for non-Euclidean structured data, have also been explored for application in the field of HSIC [34], [35]. Ding et al. [36] proposed a space spectral adaptive probability graph (SSAPG) structure that combines spectral information and spatial coordinates, and achieved good results in HSIC using graph neural networks.

B. Transformer and Vision Transformer

The Transformer architecture [37] is the most remarkable method in the field of deep learning in recent years. The core self-attention mechanism obtains the correlation between input sequences tokens through three different weights: W_k, W_v, and W_q. The calculation process is shown in Fig. 1

$$\mathrm{SA} = \mathrm{attention}(K, Q, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \quad (1)$$

Transformer was first used in natural language processing, and with the extension of the Google team, the Vision Transformer architecture [38] has emerged that can be used in the field of image processing. Due to its excellent performance in scenarios with large amounts of data, it has also appeared in many applications in the field of HSIC [39]. Neural networks may have limitations in capturing spectral pixel features, while Transformers can better capture the relationships between spectral bands within pixels, especially in processing categories with similar spectral information [18]. HSI-BERT [41] is the first to introduce the Transformer architecture into the field of hyperspectral image processing. HSI-BERT has a global receptive field that can better capture the correlation between pixels and their adjacent pixels. In [42], an attention-driven residual dense network (SGARDN) was proposed for extracting effective shallow spatial spectral features. In [10], a spectral–spatial feature tokenization transformer (SSFTT) method is proposed to capture spectral–spatial features and high-level semantic features. This approach takes into account the spectral characteristics of hyperspectral image spectra by combining and dividing the inputs of the Transformer in a certain way according to the spectral channels. Yu et al. [40] proposed a multilevel spectral–spatial transformer network (MSTNet) for HSIC.

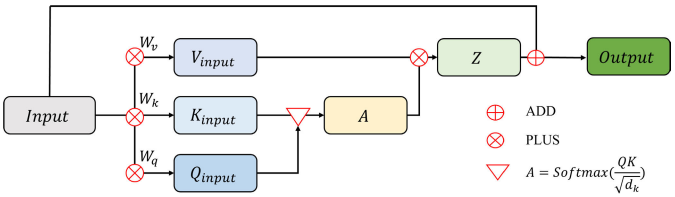


Fig. 1. Schematic of the self-attention. All its computations are based on the input.

C. Masked Image Reconstruction

Autoencoder aims to train a model that can effectively extract model features using unlabeled or semilabeled data, and it has been applied in the field of HSIC [44] [45]. Masked autoencoder refers to the use of neural networks to learn the information in data by masking or damaging a portion of the data, and restoring the original data. It was first used as a large model pretraining method for natural language processing, such as BERT [46], which uses word masking on sentences in the corpus to restore the original text and complete model training. He et al. [47] first used masked autoencoders as a method of self-supervised pretraining for image large-scale models. The model achieved classification performance similar to supervised learning models on the ImageNet dataset. Corresponding work soon appeared in the video field [48].

In recent years, this method has also been applied in the field of HSIC. Ibañez et al. [18] combined this method with Spectral Spatial Feature Token Transformer for HSIC. This approach is the first use of MAE on the field of HSIC and provides a novel approach to HSIC models. Cao et al. [19] proposed a method that combines masked autoencoders and contrastive learning loss for experimentation. Zhang et al. [49] proposed a method based on masked self-supervised network, and designed a new proxy task based on masking and reconstruction to achieve self-supervised learning.

III. Methodology

In this section, we introduce the specific methods and technical details of the cross-dataset masked self-supervised HSIC model. The method proposed in this article can effectively use multiple existing hyperspectral image datasets to complete the pretraining process of the classification model, and optimize the problem of label and channel differences in hyperspectral cross-dataset training. We will first introduce the adaptive module that improves the problem of cross-dataset channels of hyperspectral images, and then introduce the strategy and details of the 3-D masked autoencoder framework designed for hyperspectral image features and the contents of the supervised fine-tuning process. Fig. 2 presents the overall flow chart of our proposed method.

A. Data Preprocessing and Channel Adaptation Module

The hyperspectral image label is a pixel-level label, and the classification result obtained by using a single pixel as input

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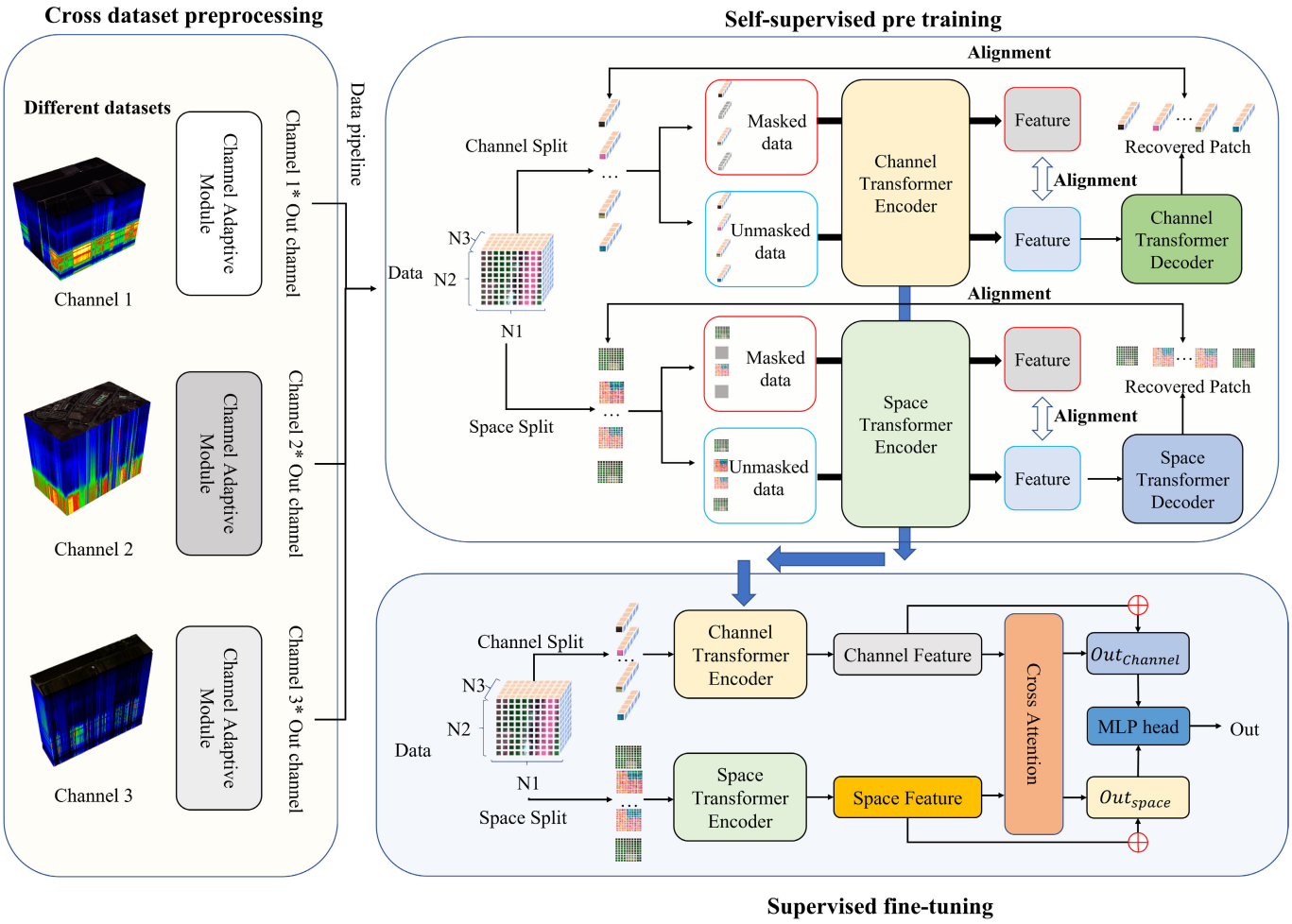


Fig. 2. Overall flowchart of the proposed method in this article. Three modules are included: channel adaptive module, self-supervised pretraining module, and supervised fine-tuning module.

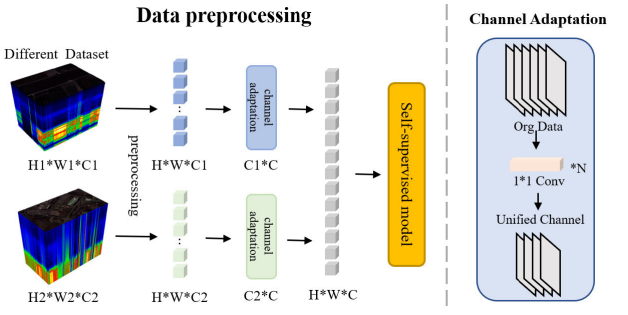


Fig. 3. Schematic of data preprocessing and channel adaptation modules. Ability to adapt hyperspectral datasets of different channels to the same data dimension as needed.

is not accurate due to the influence of noise and other factors. In the data preprocessing process of this article, we expand the hyperspectral image into a square of equal size according to the central pixel, so that the hyperspectral image data can obtain a larger receptive field on the basis of a uniform data format in the spatial direction, and also it can better adapt to the subsequent 3-D masked self-supervised learning and training processes.

Aiming at the problem of band differences in training HSIC models across datasets, the channel adaptation module we propose uses a 1*1 convolutional layer with variable parameters as the main method to adjust the number of hyperspectral channels. The 1*1 convolution kernel can adjust the data depth without changing the size of hyperspectral image data, and can effectively complete the adaptation of hyperspectral image datasets with different channel numbers. Each hyperspectral dataset with different channel numbers can incorporate self-supervised learning into the training process using a unique channel adaptation module. This module can carry out an end-to-end training process and adjust the weight parameters of each channel by itself. Although there are differences in the channel bands between hyperspectral image datasets, there is a link between the spectra. The channel adaptation module uses the link between the spectra to complete the channel matching across datasets.

As shown in Fig. 3, for two datasets with channel numbers $C1$ and $C2$, after dividing into $H * W$ data blocks and after passing through the corresponding channel adaptive module for the respective channel number, they are unified into data blocks with channel number C, and the shape of the data block is still $H * W$. Let the input shape be $H_{\text{in}} * W_{\text{in}} * C_{\text{in}}$, the output

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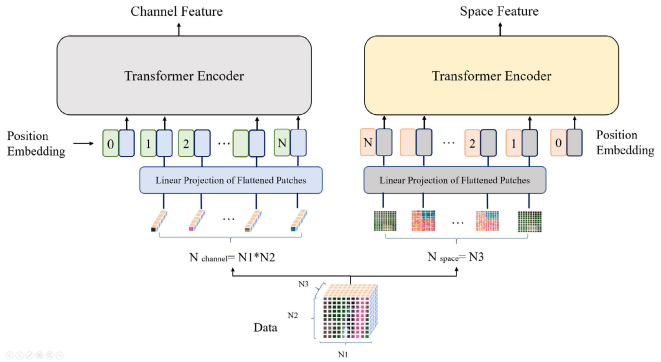


Fig. 4. Schematic of the two-branch Transformer structure.

shape $H_{\text{out}} * W_{\text{out}} * C_{\text{out}}$, and the convolution kernel $1*1*k$. The input–output relationship is shown in the following formula:

$$\begin{aligned} H_{\text{in}} &= H_{\text{out}} \\ W_{\text{in}} &= W_{\text{out}} \\ C_{\text{out}} &= k. \end{aligned} \tag{2}$$

The abovementioned data preprocessing process and channel adaptation module can effectively solve the problem of channel differences in training classification models across hyperspectral datasets. This scheme allows the use of any number of hyperspectral datasets with any number of channels for pretraining and fine-tuning.

B. Dual-Branch Transformer and Cross Attention

In the previous background, we have introduced and explained the attention mechanism and Vision Transformer architecture. However, in the field of hyperspectral image classification, the application of Vision Transformers for simultaneously acquiring spatial and spectral information is not yet fully mature. To effectively utilize both the spatial and spectral information in hyperspectral images, we propose a spatial–spectral Transformer that fuses a dual-branch structure with cross-attention. A schematic diagram of the proposed two-branch Transformer structure is shown in Fig. 4.

In the traditional Vision Transformer architecture, the input image data is cut into different image blocks unfolded as a single sequence of data, and for hyperspectral image data, the data processing is also different from the traditional RGB image data because it usually has more number of channels. Considering the need to use both spatial and spectral information of the data, the spatial and spectral correlations can be extracted separately according to the information that needs to be emphasized when designing the two-branch Transformer. In the spatial information extraction branch, the original image is sliced according to the channels and each spectral image is used as a unit of the sequence to obtain the sequence of the number of spectral channels, focusing on the spatial correlation. In the spectral information extraction branch, the spectra of individual pixels can be used as a unit of the sequence to obtain the sequence of the number of image pixels, emphasizing the spectral correlation.

After passing through the two-branch Transformer, the spatial and spectral features of the original image patch can be acquired, and in order to fuse the two features more well, the proposed method introduces the cross attention as the measure of feature fusion. Cross attention is a type of attention mechanism mostly used for correlation acquisition between two different sequences. The use of cross attention replaces the original single-ended feature fusion approach, ensures the adequacy of feature fusion, and realizes the combination of spectral and spatial feature information. The flow is shown in Fig. 5. Specifically, when two different sequences $\text{Feature}_{\text{space}}$ and $\text{Feature}_{\text{channel}}$ enter the cross-attention layer, they will acquire their own unique K, Q, and V according to the learnable weight matrices W_k, W_q, and W_v as in the case of the self-attention mechanism.

$$\begin{aligned} Q_{\text{space}} &= \text{Feature}_{\text{space}} W_q \\ K_{\text{space}} &= \text{Feature}_{\text{space}} W_k \\ V_{\text{space}} &= \text{Feature}_{\text{space}} W_v. \end{aligned} \tag{3}$$

And in the subsequent computation process, the Q matrices between two different sequences will be exchanged with each other, and Q_{space} will be computed with K_{channel} and V_{channel}, so as to realize the information exchange and fusion between the sequences. This approach realizes interactive learning between spatial and spectral features, and after the computation, the output cross features will be added to the original spatial and spectral features using residual links to way the features after the cross-attention layer for information loss or feature degradation

$$\begin{aligned} A_{\text{space}*\text{channel}} &= \text{softmax}\left(\frac{Q_{\text{channel}} K_{\text{space}}^T}{\sqrt{d_k}}\right) \\ A_{\text{channel}*\text{space}} &= \text{softmax}\left(\frac{Q_{\text{space}} K_{\text{channel}}^T}{\sqrt{d_k}}\right) \end{aligned} \tag{4}$$

$$\begin{aligned} \text{Out}_{\text{space}} &= A_{\text{space}*\text{channel}} V_{\text{space}} + \text{Feature}_{\text{spacel}} \\ \text{Out}_{\text{channel}} &= A_{\text{channel}*\text{space}} V_{\text{channel}} + \text{Feature}_{\text{channel}}. \end{aligned} \tag{5}$$

After the above calculation, it is possible to add spectral correlation features to the spatial feature group under the premise of maintaining the original spatial features, and vice versa. The design of the dual-branch spatial Transformer joint cross-attention mechanism makes full use of the characteristics of hyperspectral data itself, realizes a unique feature extraction and fusion method, and is able to effectively use the information in the hyperspectral data to achieve the purpose of feature extraction.

C. Masked Self-Supervised Learning Method Based on Context

Currently, masked autoencoder is a very effective self-supervised learning method, in which the commonly used masked autoencoder consists of an encoder and a decoder, and this method is usually a mixture of two tasks, the encoder's task can be viewed as the learning of the feature representation, while the decoder's task is the recovery of the masked image blocks. In MAE architectures, these two tasks are relatively undifferentiated, which makes the feature representation learned by the encoder limit the performance of the decoder in reconstructing the original image. Current approaches lack the intercorrelation of features between the

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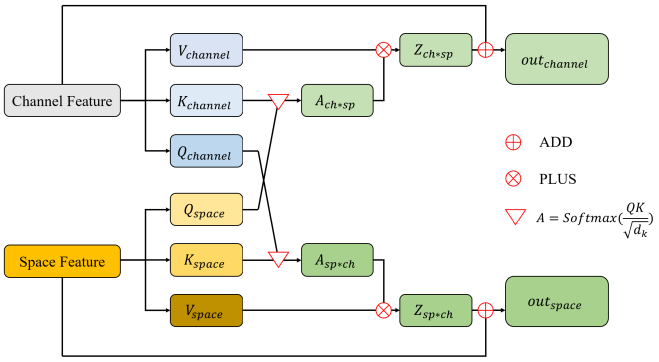


Fig. 5. Schematic of cross attention.

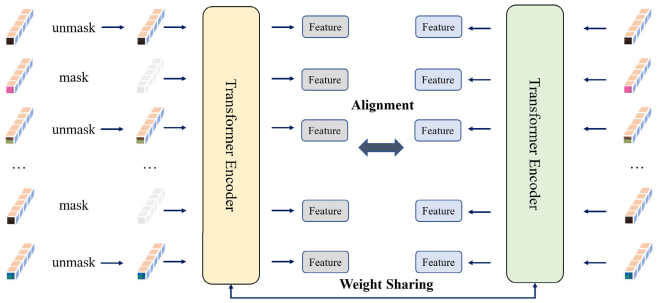


Fig. 6. Schematic of the encoder architecture for context masked autoencoder.

visible image block and the intermediate features of the masked image block. In order to obtain a better correlation between the feature extraction of masked and explicit image blocks, a self-supervised learning method for contextual features based on intermediate features is proposed in this chapter, and its flowchart is shown in Fig. 6.

In the data processing stage, the contextual feature self-supervised learning based on intermediate features adopts the masked processing similar to the masked autoencoder. Since the method used in this stage adopts the two-branch spatial–spectral Transformer mentioned in Section III-B as the backbone network, the division of the input data is divided into two kinds according to the spatial information and according to the spectral information, and the division according to the spectral information is used as an example for illustration here. For the input data divided by spectral information, each piece of data can be regarded as the spectrum of a single pixel, and all the data will be randomly divided into two groups of visible $\text{Patch}_{\text{unmask}}$ and mask $\text{Patch}_{\text{mask}}$. In the self-supervised training phase, as an encoder the Transformer receives the visible data blocks, fills in the corresponding positional encoding, and obtains the feature representation of the visible data blocks based on the visible data and positional information, and tries to predict the feature representation of the masked data. Based on the construction of the auxiliary task, the encoder will take all the data blocks as inputs and get the feature representation of all the data as the actual label, and the feature representation of the masked data will align the input features of all the data blocks to try to obtain a consistent feature representation. Compared to the masked self-encoder which performs direct recovery of the original image, this semantic contextual

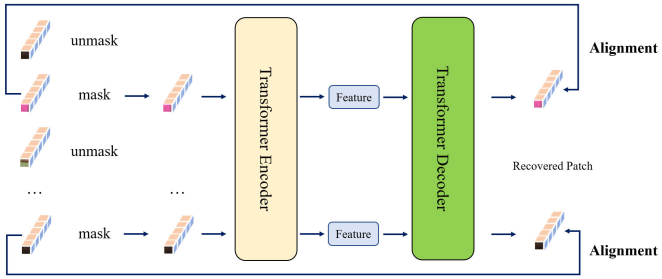


Fig. 7. Schematic of the decoder architecture for context masked autoencoder.

feature-based encoder derives the feature representation of the masked image from the feature representation of the visible image block, whose semantic features are in the same feature space. And because the features learned by the encoder will be further applied to the subsequent processing stages, the feature representation task will be done by the encoder, while the decoder is not involved in the feature representation task, which can circumvent the performance limitations imposed by the decoder

$$\begin{aligned} \text{Out}_{\text{masked}} &= \text{Encoder}(\text{Patch}_{\text{unmask}}) \\ \text{Out}_{\text{label}} &= \text{Encoder}(\text{Patch}_{\text{unmask}}\text{Patch}_{\text{masked}}). \end{aligned} \tag{6}$$

After the encoder stage, what is obtained is still only the feature representation of the image block, and the decoder decoding operation is still needed to get the final image recovery. Because the encoder stage has taken the features as a means of recovery to get the encoder with feature representation capability, so in the decoder does not need to consider the feature acquisition, but only need to consider the mask recovery problem. In the decoder architecture, as shown in Fig. 7, the backbone network of the decoder still adopts the two-branch Transformer architecture for the different recovery of spectral and spatial information, but compared with the mask self-supervised learning decoder, the input of the decoder here adopts only the feature representation of the masked image data instead of all of the feature representations, and utilizes the masked original image as the label for decoding and recovery. The decoder is only involved in the image recovery stage of the mask without affecting the feature extraction capability of the encoder.

In the construction of the loss function, the mean square error loss is used for both feature and image recovery, considering that both feature representation and image recovery need to use the original data or features as labels. Assuming that the input of the spectral representation is $\text{Patch}_{\text{channel}}$, the masked output after the encoder is $\text{Out}_{\text{masked}}$, the complete output after the encoder is $\text{Out}_{\text{label}}$, and the output of the decoder is $\text{Patch}_{\text{recover}}$, the loss function can be considered as follows:

$$\text{Loss} = \ell(\text{Out}_{\text{masked}}, \text{Out}_{\text{label}}) + \lambda\ell(\text{Patch}_{\text{recover}}, \text{Patch}_{\text{channel}}). \tag{7}$$

D. Supervised Fine-Tuning

In order to enable the self-supervised pretrained model to be effectively applied to HSIC tasks, it is necessary to perform

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TABLE I
INDIAN PINES DATASET DETAILS

Class NO.	Class Name	Training	Testing
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture	48	435
6	Grass-trees	73	657
7	Grass-pasture-mowed	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	245	2210
12	Soybean-clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	9	84
	Total	1024	9225

TABLE II
PAVIAU DATASET DETAILS

Class NO.	Class Name	Training	Testing
1	Asphalt	663	5968
2	Meadows	1865	16784
3	Gravel	210	1889
4	Trees	306	2758
5	Painted metal sheets	134	1211
6	Bare Soil	503	4526
7	Bitumen	133	1197
8	Self-Blocking Bricks	368	3314
9	Shadows	95	852
	Total	4277	38499

TABLE III
WHU-HI-HANCHUAN DATASET DETAILS

Class NO.	Class Name	Training	Testing
1	Strawberry	4473	40262
2	Cowpea	2275	20478
3	Soybean	1029	9258
4	Sorghum	535	4818
5	Water spinach	120	1080
6	Watermelon	453	4080
7	Greens	590	5313
8	Trees	1798	16180
9	Grass	947	8522
10	Red roof	1052	9464
11	Gray roof	1691	15220
12	Plastic	368	3311
13	Bare soil	912	8204
14	Road	1856	16704
15	Bright object	114	1022
16	Water	7540	67861
	Total	25753	231777

TABLE IV
WHU-HI-LONGKOU DATASET DETAILS

Class NO.	Class Name	Training	Testing
1	Corn	3451	31060
2	Cotton	837	7537
3	Sesame	303	2728
4	Broad-leaf soybean	6321	56891
5	Narrow-leaf soybean	415	3736
6	Rice	1185	10669
7	Water	6706	60350
8	Roads and houses	713	6411
9	Mixed weed	523	4706
	Total	20454	184088

corresponding supervised fine-tuning on the pretrained model. In the supervised learning stage, the model uses the same architecture and model parameters as the encoder in the self-supervised learning stage, and the corresponding fully connected layer is added as the classification head according to the number of dataset categories used for fine-tuning. In the supervised learning stage, the data input is all unmasked hyperspectral image data blocks. Since the training task in the supervised fine-tuning phase is a classification task, and the label used is the category label corresponding to the central pixel of the tile, we use the cross-entropy loss function for training

$$\text{CrossEntropy Loss} = -\frac{1}{N}\sum_{i}\sum_{c=1}^{M} y_{ic}\log(p_{ic}) \tag{8}$$

where M represents the number of categories, y_{ic} indicates whether the data prediction is correct, and p_{ic} represents the probability that i belongs to c. The supervised fine-tuning stage also fully realizes the cross-dataset classification application. Whether it is a hyperspectral dataset that participates in pretraining or a hyperspectral dataset that does not participate in pretraining, supervised fine-tuning can be used to obtain the final classification model.

IV. EXPERIMENTS AND ANALYSIS

In this section, we conduct extensive experiments on multiple hyperspectral datasets for the proposed method and compare the performance of the self-supervised learning method with baseline and state-of-the-art supervised methods. First, the hyperspectral dataset will be used and the experimental settings and implementation details will be introduced. Then the performance of the model and the evaluation and comparison of other methods will be shown. Finally, the parameter selection and training of the model will be analyzed.

A. Introduction of Datasets

1) Indian Pines Dataset: The Indian Pines dataset was one of the first datasets used for HSIC tasks, collected using the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) at and around the Purdue University Agronomy Farm in northwest West Lafayette. The image size of the dataset is 145 × 145 pixels, including 220 band data of wavelengths from 400 to 2500 nm, of which only 10 249 pixels are surface object pixels, and the remaining 10 776 pixels are background pixels. The annotated ground object pixels are classified into 16 categories with an unbalanced number of samples. The details of the training and testing data samples for each class are showed in Table I.

2) Pavia University Dataset: The University of Pavia dataset is part of the hyperspectral imagery collected in the city of Pavia by the airborne Reflective Optical Spectral Imaging System (ROSIS-03). The image size of the dataset is 610 × 340 pixels, including 115 band data of wavelengths from 430 to 860 nanometers, of which 42 776 pixels are

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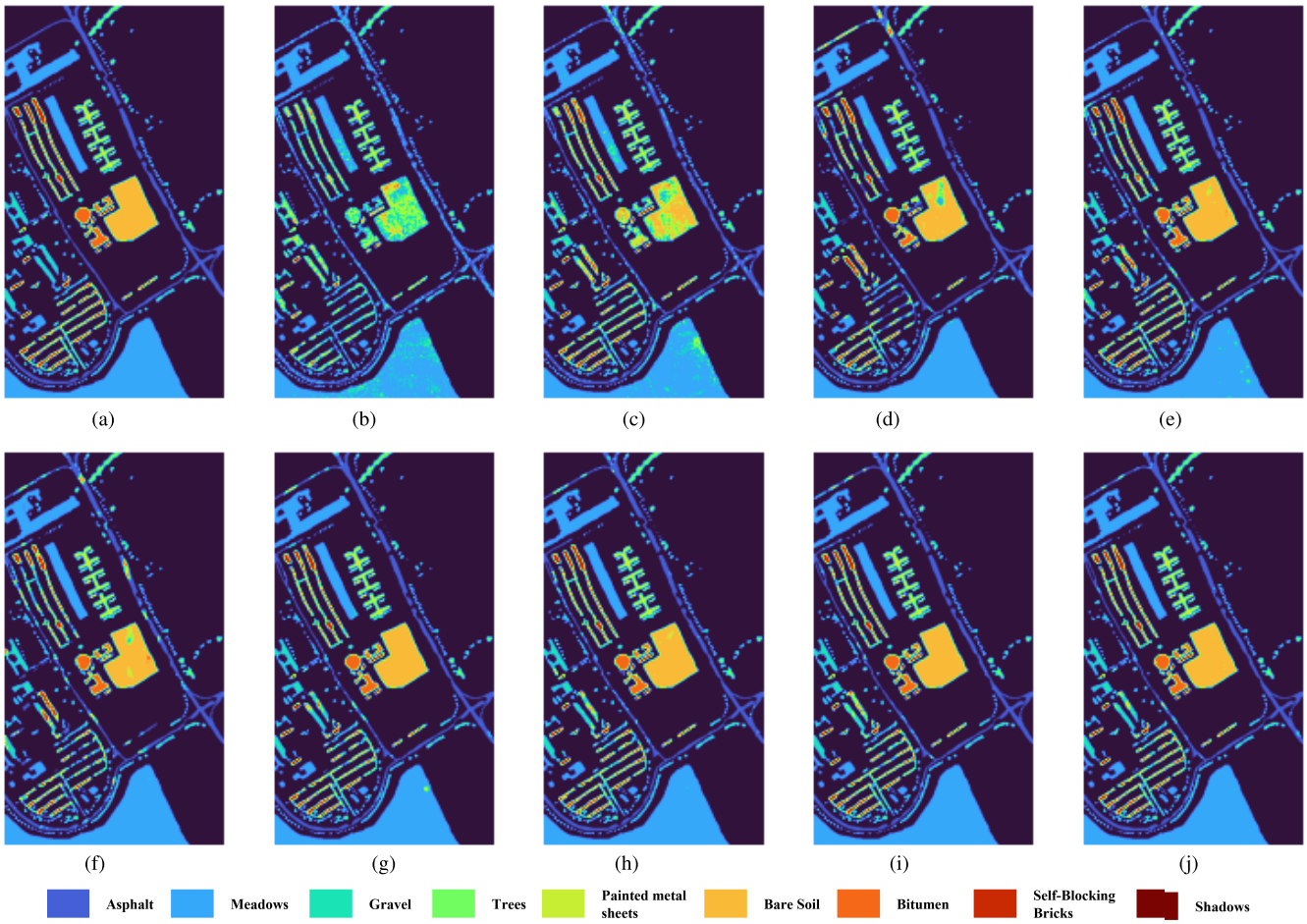


Fig. 8. Hyperspectral classification results of the original image labels and each comparison method on the PaviaU dataset. (a) GT. (b) KNN. (c) SVM. (d) 3D-CNN. (e) SSTN. (f) MAEST. (g) ViT. (h) HybridSN. (i) HSIMAE. (j) Ours.

TABLE V
WHU-HI-HONGHU DATASET DETAILS

Class NO.	Class Name	Training	Testing
1	Red roof	1404	12637
2	Road	351	3161
3	Bare soil	2182	19639
4	Cotton	16328	146957
5	Cotton firewood	622	5596
6	Rape	4456	40101
7	Chinese cabbage	2410	21693
8	Pakchoi	405	3649
9	Cabbage	1082	9737
10	Tuber mustard	1239	11155
11	Brassica parachinensis	1102	9913
12	Brassica chinensis	895	8059
13	Small Brassica chinensis	2251	20256
14	Lactuca sativa	736	6620
15	Celtuce	100	902
16	Film covered lettuce	726	6536
17	Romaine lettuce	301	2709
18	Carrot	322	2895
19	White radish	871	7841
20	Garlic sprout	349	3137
21	Broad bean	133	1195
22	Tree	404	3636
	Total	38669	348024

surface object pixels. The details of the training and testing data samples for each class are shown in Table II.

3) WHU-Hi-HanChuan Dataset: The WHU-Hi-HanChuan dataset was acquired from 17:57 to 18:46 on June 17, 2016, in Hanchuan, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform. During the data collection, the weather was clear and cloudless, the temperature was about 30 °C, and the relative air humidity was about 70%. The study area is a rural–urban fringe zone with buildings, water, and cultivated land, which contains seven crop species: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. The UAV flew at an altitude of 250 m, the size of the imagery is 1217 × 303 pixels, there are 274 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is about 0.109 m. Notably, since the WHU-Hi-HanChuan dataset was acquired during the afternoon when the solar elevation angle was low, there are many shadow-covered areas in the image.

4) WHU-Hi-LongKou Dataset: The WHU-Hi-LongKou dataset was acquired from 13:49 to 14:37 on July 17, 2018, in Longkou Town, Hubei province, China, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. During the data collection, the weather was clear and cloudless, the

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TABLE VI

ACCURACY OF CLASSIFICATION RESULTS AND EVALUATION METRICS FOR COMPARATIVE METHODS ON THE PAVIAU DATASET

Class	KNN	SVM	3D CNN	SSTN	MAEST	ViT	HybridSN	HSIMAE	Ours
1	70.62	90.19	96.10	93.52	85.43	85.73	96.65	95.59	98.70
2	82.97	92.41	96.15	99.37	99.00	99.37	98.16	99.81	99.86
3	60.40	82.68	77.29	82.28	75.48	98.26	85.28	90.72	97.63
4	98.06	95.21	96.48	96.93	96.57	99.89	95.40	95.66	98.84
5	100.00	100.00	99.59	99.38	99.26	99.92	99.42	99.67	99.75
6	80.34	89.16	97.39	98.58	97.88	98.82	97.66	99.61	99.91
7	53.69	79.18	78.95	92.14	80.20	97.44	85.80	96.95	99.33
8	62.54	84.54	93.57	81.95	73.49	87.01	91.28	90.25	93.35
9	98.84	100.00	98.94	92.48	45.15	98.58	98.83	94.95	98.10
OA(%)	79.05	90.87	94.83	95.46	91.77	95.88	96.16	97.36	98.87
AA(%)	72.60	88.59	92.72	92.96	83.61	94.20	94.27	95.91	98.22
Kappa(%)	77.76	90.70	93.10	95.48	91.47	95.78	94.86	97.37	98.87

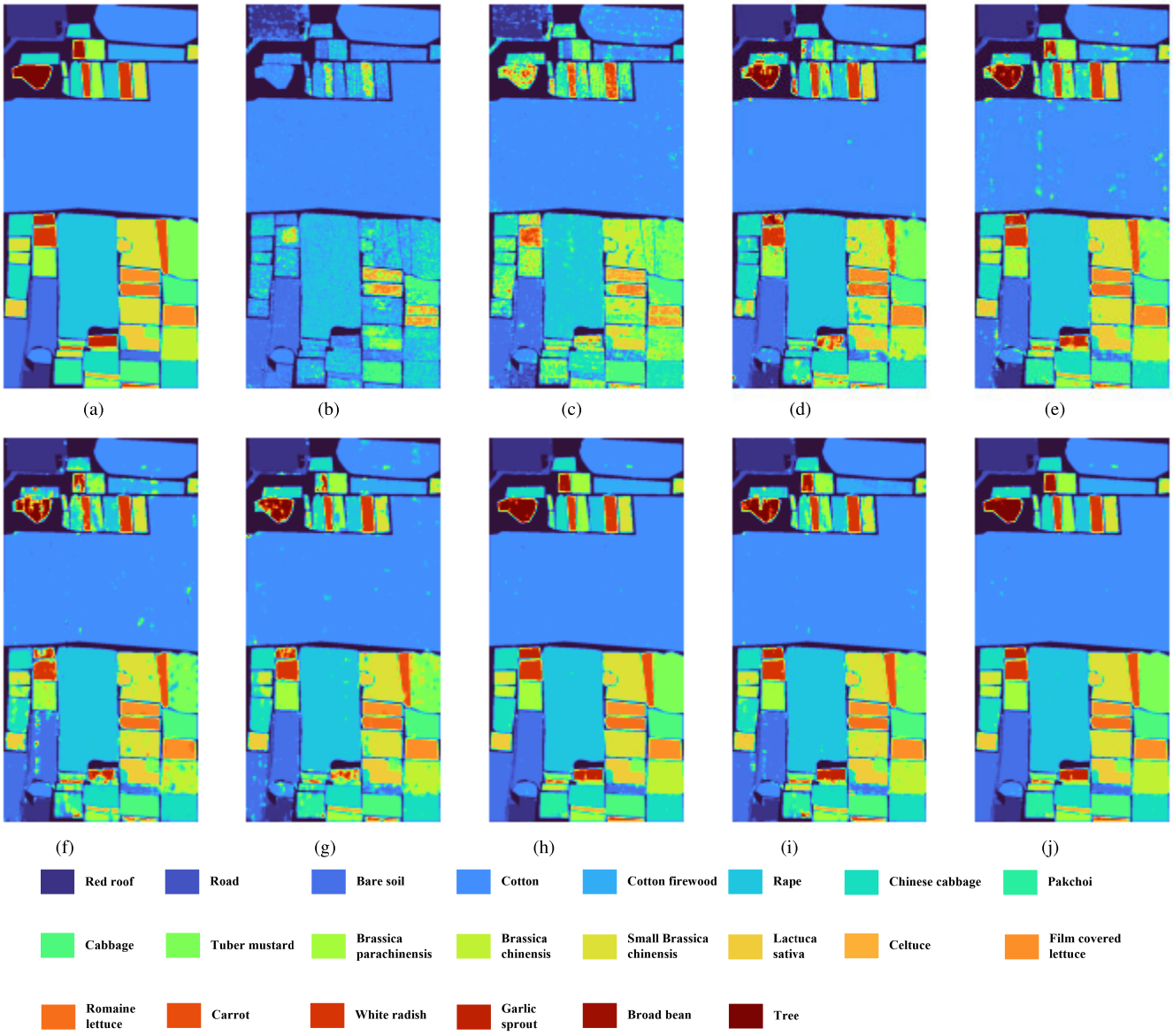


Fig. 9. Hyperspectral classification results of the original image labels and each comparison method on the WHU-Hi-HongHu dataset. (a) GT. (b) KNN. (c) SVM. (d) 3D-CNN. (e) SSTN. (f) MAEST. (g) ViT. (h) HybridSN. (i) HSIMAE. (j) Ours.

temperature was about 36 °C, and the relative air humidity was about 65%. The study area is a simple agricultural scene, which contains six crop species: corn, cotton, sesame, broadleaf soybean, narrow-leaf soybean, and rice. The UAV flew

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TABLE VII
ACCURACY OF CLASSIFICATION RESULTS AND EVALUATION METRICS FOR COMPARATIVE METHODS ON THE WHU-HI-HONGHU DATASET

Class	KNN	SVM	3D CNN	SSTN	MAEST	ViT	HybridSN	HSIMAE	Ours
1	70.59	86.89	94.01	94.28	95.81	97.84	98.46	97.4	99.04
2	27.30	48.61	55.02	71.34	76.97	87.22	75.45	88.87	90.01
3	65.64	81.14	85.32	90.36	92.96	94.98	95.61	95.84	97.96
4	77.70	95.99	96.51	97.15	98.5	98.48	99.4	98.13	99.63
5	1.38	27.30	72.42	83.24	85.63	84.61	88.93	78.73	96.52
6	68.00	86.46	96.15	97.87	98.55	98.6	98.41	98.81	99.3
7	38.47	67.50	75.22	85.11	84.75	92.5	95.77	94.1	97.54
8	3.86	12.39	43.93	68.67	72.15	87.5	93.72	87.53	96.21
9	79.85	93.73	94.6	97.36	96.94	97.91	97.91	98.14	99.31
10	15.96	54.29	79.02	87.33	80.5	92.9	93.73	94.97	97.53
11	8.02	43.26	68.58	82.57	88.28	88.1	91.88	91.95	96.62
12	10.92	33.64	59.73	76.89	72.91	91.23	92.17	92.8	95.3
13	21.24	58.98	75.51	85.44	87.47	91.7	94.22	92.68	96.63
14	28.55	60.04	75.06	88.93	86.37	91.74	92.83	95.31	97.35
15	2.18	17.14	59.67	87.46	93.74	94.02	87.59	94.56	96.92
16	71.07	75.11	90.65	93.43	96.86	97.69	97.3	98.14	99.36
17	37.82	39.07	71.18	86.86	90.89	95.59	92.21	96.94	99.06
18	4.10	29.32	75.97	90.09	91.58	93.51	89.72	90.37	96.07
19	35.08	72.59	87.82	92.5	91.29	91.34	91.73	95.63	96.95
20	1.45	41.96	69.81	75.86	66.39	86.62	89.22	88.67	95.36
21	0.30	1.68	0.66	64.04	61.06	84.24	83.35	85.15	95.19
22	0.62	44.54	66.4	83.61	86.35	88.9	95.38	91.16	97.82
OA(%)	63.60	80.30	88.08	92.61	93.32	95.77	96.76	96.24	98.52
AA(%)	30.46	53.26	72.42	85.47	86.18	92.14	92.5	92.99	97.08
Kappa(%)	57.11	78.88	87.56	92.42	93.31	95.77	96.77	96.18	98.51

TABLE VIII
ACCURACY OF CLASSIFICATION RESULTS AND EVALUATION METRICS FOR COMPARATIVE METHODS ON THE WHU-HI-HANCHUAN DATASET

Class	KNN	SVM	3D CNN	SSTN	MAEST	ViT	HybridSN	HSIMAE	Ours
1	70.87	88.20	96.66	97.28	98.52	98.78	98.44	98.14	98.48
2	60.84	68.83	92.99	94.73	97.26	97.28	97.64	96.54	97.76
3	47.30	69.88	88.85	94.76	97.51	96.03	98.06	97.82	98.54
4	81.29	87.94	92.38	98.33	98.92	98.59	99.48	99.15	99.58
5	1.12	10.79	81.77	85.04	87.13	89.48	95.11	90.74	94.84
6	12.92	18.02	83.35	83.19	89.72	87.89	91.84	90.19	92.53
7	27.56	68.27	89.86	91.83	91.03	91.85	96.14	94.26	96.33
8	36.86	66.82	83.83	95.53	95.92	95.60	97.87	96.82	97.97
9	30.11	55.93	88.90	93.16	89.91	93.28	97.09	95.62	97.42
10	62.26	92.62	82.15	97.83	95.41	94.63	98.10	97.89	98.25
11	44.38	83.61	91.11	96.29	95.73	95.27	97.61	96.42	97.63
12	3.72	26.61	73.78	89.06	92.13	94.78	93.82	89.57	94.32
13	11.08	52.19	81.18	82.28	90.29	92.48	90.26	87.06	90.54
14	64.33	80.67	90.83	93.68	93.93	95.76	96.73	94.91	96.78
15	46.55	56.80	76.31	88.73	84.13	84.90	93.77	91.80	94.12
16	85.85	98.77	99.36	99.65	99.50	99.45	99.70	99.44	99.74
OA(%)	64.92	81.97	92.80	96.01	96.70	96.95	97.91	97.05	98.03
AA(%)	42.94	64.12	87.08	92.59	93.57	94.13	96.35	94.77	96.55
Kappa(%)	61.68	81.15	92.74	95.98	96.70	96.95	97.90	97.04	98.02

at an altitude of 500 m, the size of the imagery is 550 × 400 pixels, there are 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is about 0.463 m.

5) WHU-Hi-HongHu Dataset: The WHU-Hi-HongHu dataset was acquired from 16:23 to 17:37 on November 20, 2017, in Honghu City, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro UAV platform. During the data collection, the weather was cloudy, the temperature was about 8 °C, and the relative air humidity was about 55%. The experimental area is a complex agricultural scene with many classes of crops, and different cultivars of the same crop are also planted in the region, including Chinese cabbage and cabbage, and Brassica chinensis and small Brassica chinensis. Notably, the region is planted with different cultivars of the same crop type; for example, Chinese cabbage/cabbage and Brassica chinensis/small Brassica chinensis. The UAV flew at an altitude of 100 m, the size of the imagery is 940 × 475 pixels, there are 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is about 0.043 m.

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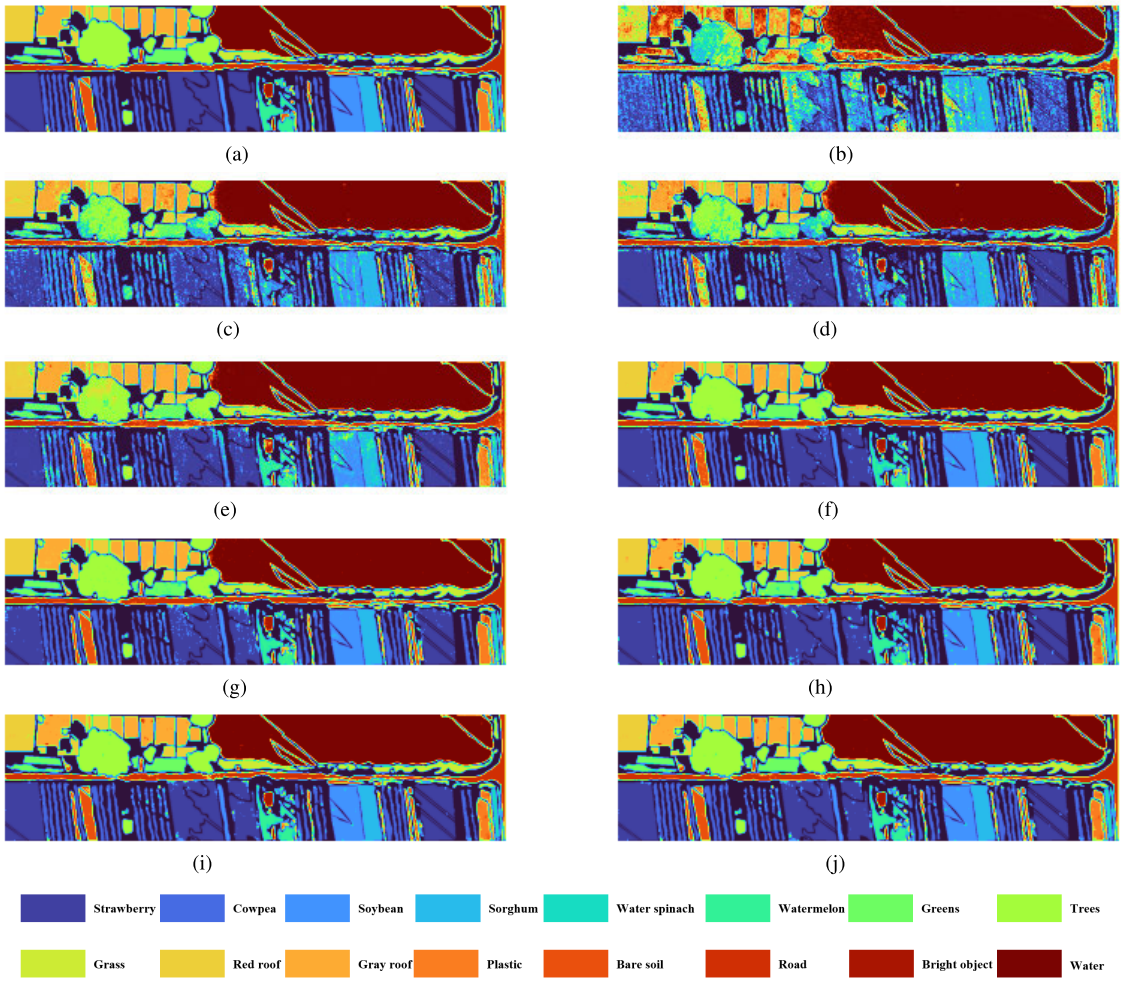


Fig. 10. Hyperspectral classification results of the original image labels and each comparison method on the WHU-Hi-Hanchuan dataset. (a) GT. (b) KNN. (c) SVM. (d) 3D-CNN. (e) SSTN. (f) MAEST. (g) ViT. (h) HybridSN. (i) HSIMAE. (j) Ours.

B. Experiment Settings

1) Implementation: The implementation of the self-supervised model pretraining and supervised fine-tuning was designed and executed using PyTorch in Python 3.9.7 on a windows machine with 12th Gen Intel[1] Core[2] i5-12400F processor with 16-GB RAM with an NVIDIA GeForce GTX 3060 12 GB graphics processing unit (GPU). Due to computational limitations, we executed the masked self-supervised reconstruction and supervised classification separately. Specifically, for classification training, we used a batch size of 64 with the Adam optimizer. A learning rate of 5e−4 decaying with a factor of 0.9 every 1/10 of the epochs was used, which were set to 1000.

Tables I–V in the dataset introduction section show the datasets we used and how they were divided. All datasets were divided in the ratio of 10% training set and 90% test set. In the pretraining phase of the self-supervised model, we used all the datasets in the unlabeled training set as the pretraining data. For all supervised models and the fine-tuning phase of the pretrained models, we used labeled data from each dataset for model training. In order to minimize the impact of data on the experimental results, we used the same training and test sets for training different models, while the training and test sets taken on multiple independent repetitions of the experiments were randomly divided.

The evaluation metrics used in this article include the following: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. The final classification results can be grouped into four categories: true positive (TP) examples, false positive (FP) examples, true negative (TN) examples, and false negative (FN) examples. The formula of the evaluation index is as follows:

$$\begin{aligned} \mathrm{AA} &= \frac{\mathrm{TN}+\mathrm{TF}}{\mathrm{TN}+\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \\ \mathrm{OA} &= \frac{1}{N} * \sum(\mathrm{OA}) \\ \mathrm{Kappa} &= \frac{\mathrm{OA}-\mathrm{AA}}{1-\mathrm{AA}}. \end{aligned} \tag{9}$$

2) Compared Methods: In order to show the performance and superiority of the method, we compare the performance of the proposed method with many representative methods existing in the field of HSIC, and then introduce the comparison method used and the related parameter settings. These methods

[1]Registered trademark.
[2]Trademarked.

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TABLE IX
MODEL CLASSIFICATION RESULTS COMPARISON ON THE INDIAN PINE DATASET

Class	Complete Model			Ablation Experimental		
	precision	recall	f1score	precision	recall	f1score
1	90.48	92.68	91.57	91.11	100.00	95.35
2	97.56	93.23	95.34	98.69	93.77	96.17
3	91.68	98.80	95.10	95.31	97.86	96.57
4	97.51	92.02	94.69	98.09	96.24	97.16
5	95.72	97.70	96.70	98.86	99.54	99.20
6	98.78	98.93	98.86	98.04	98.78	98.41
7	100.00	64.00	78.05	100.00	100.00	100.00
8	99.31	100.00	99.65	100.00	100.00	100.00
9	100.00	44.44	61.54	100.00	61.11	75.86
10	96.26	94.06	95.14	95.16	96.69	95.92
11	95.38	98.10	96.72	96.04	98.69	97.34
12	97.80	91.39	94.48	97.65	93.45	95.50
13	99.45	97.84	98.64	97.85	98.38	98.11
14	97.49	98.95	98.21	97.85	100.00	98.91
15	96.78	95.39	96.08	98.43	90.20	94.14
16	93.98	92.86	93.41	98.72	91.67	95.06
OA			96.42			97.20
AA	96.76	90.65	92.76	97.61	94.77	95.86
Kappa	96.48	96.42	96.39	97.24	97.20	97.18

TABLE X
MODEL CLASSIFICATION RESULTS COMPARISON ON THE WHU-HI-LONGKOU DATASET

Class	Complete Model			Ablation Experimental		
	precision	recall	f1score	precision	recall	f1score
1	99.87	99.87	99.87	99.86	99.93	99.89
2	99.67	99.75	99.71	99.75	99.99	99.87
3	99.89	99.93	99.91	99.85	99.89	99.87
4	99.82	99.92	99.87	99.89	99.80	99.84
5	99.28	99.12	99.20	98.86	99.46	99.16
6	99.77	99.89	99.83	99.79	99.93	99.86
7	99.91	99.99	99.95	99.93	99.97	99.95
8	99.19	95.40	97.26	98.33	98.47	98.40
9	95.74	98.34	97.02	98.84	97.60	98.21
OA			99.71			99.78
AA	99.24	99.13	99.18	99.46	99.45	99.45
Kappa	99.71	99.71	99.71	99.78	99.78	99.78

include traditional classification models and neural network models, and we use KNN, SVM, 3D-CNN, MAEST, ViT, HybridSN, SSTN, and HSIMAE as the comparison method of classification models, where KNN and SVM are relatively traditional classification models, and other methods are neural network classification models.

The specific parameters of each model are detailed below. For the KNN model, the selected value of K is the main factor, and we set the value of K to 5. The SVM model is completed using the SKlearn library, and the selected kernel function is a linear kernel function. The model construction of 3D-CNN consists of two layers of 3-D convolutional layers and two layers of full connection. HybridSN refers to the model architecture used by the author in this article, and the classifier model used includes three 3-D convolutional layers and one 2-D convolutional layer, and adds three fully connected layers to output classification results. When using the ViT model, we used six layers of Transformer and one fully connected layer to complete the model building.

TABLE XI
MODEL CLASSIFICATION RESULTS COMPARISON ON THE WHU-HI-HANCHUAN DATASET

Class	Complete Model			Ablation Experimental		
	precision	recall	f1score	precision	recall	f1score
1	98.38	98.82	98.60	97.53	98.76	98.14
2	98.75	96.69	97.71	95.08	98.05	96.54
3	78.25	97.52	86.83	98.14	97.50	97.82
4	99.96	54.73	70.73	99.25	99.05	99.15
5	99.62	98.24	98.93	93.17	88.43	90.74
6	89.14	76.62	82.40	87.01	93.60	90.19
7	95.85	98.29	97.05	97.54	91.19	94.26
8	92.95	97.45	95.15	97.02	96.61	96.82
9	95.94	94.92	95.43	94.93	96.32	95.62
10	99.56	97.73	98.63	98.45	97.34	97.89
11	98.14	94.15	96.10	96.52	96.33	96.42
12	99.63	88.43	93.70	92.09	87.19	89.57
13	90.96	92.89	91.92	86.48	87.64	87.06
14	97.69	98.81	98.25	98.13	91.89	94.91
15	95.82	98.63	97.20	98.43	86.01	91.80
16	98.93	99.91	99.42	99.18	99.69	99.44
OA			96.63			97.05
AA	95.60	92.74	93.63	95.56	94.10	94.77
Kappa	96.84	96.63	96.53	97.07	97.05	97.04

For the HSIMAE, which incorporates more detailed components, the self-supervised learning stage employs a six-layer Transformer architecture that is entirely consistent with the ViT model. The decoder uses a single Transformer layer for decoding. The input patch size is set to 15×15 for segmentation, and all patches are adjusted to a depth of 30 channels via a channel adaptive layer. The resulting image blocks are divided into segments of size $5 \times 5 \times 6$.

In the self-supervised pretraining, we construct two cases for experiments, including the use of the first four datasets for pretraining and all five datasets for pretraining, the data presented in the comparison experiment phase are the training results of pretraining using four datasets, and we will focus on comparing the impact of whether or not dataset data is used in pretraining on the classification results in the ablation experiment phase.

C. Compare Experimental Results

The experimental results in this chapter show the performance of different comparative methods and our proposed method on the Pavia University, WHU-Hi-HongHu and WHU-Hi-Hanchuan datasets. Table VI shows the experimental

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TABLE XII
MODEL CLASSIFICATION RESULTS COMPARISON ON THE WHU-HI-HONGHU DATASET

Class	HSIMAE + ViT		HSIMAE + CATrans		ContextMAE + ViT		ContextMAE+CATrans	
	Precision	F1-score	Precision	F1-score	Precision	F1-score	Precision	F1-score
1	97.10	97.40	98.77	98.73	99.15	98.86	99.18	99.04
2	83.99	88.87	95.06	90.36	86.55	89.93	87.35	90.01
3	94.30	95.84	95.82	95.87	98.45	97.84	98.52	97.96
4	99.86	98.13	99.68	99.28	99.76	99.54	99.80	99.63
5	69.21	78.73	85.10	89.80	93.75	95.70	94.59	96.52
6	98.67	98.81	96.90	98.33	99.18	99.13	99.25	99.30
7	90.77	94.10	93.89	95.10	98.00	97.62	98.08	97.54
8	78.65	87.53	91.18	91.92	93.01	95.51	94.33	96.21
9	97.01	98.14	99.37	98.17	98.77	99.24	98.97	99.31
10	95.18	94.97	97.51	92.52	96.89	96.96	97.28	97.53
11	91.36	91.95	92.66	93.57	96.97	95.95	97.88	96.62
12	93.13	92.80	81.02	87.19	95.05	94.61	95.95	95.30
13	90.25	92.68	94.79	92.60	95.30	96.00	95.76	96.63
14	97.24	95.31	92.78	93.39	94.74	96.34	95.95	97.35
15	94.46	94.56	92.90	93.63	94.24	95.83	96.01	96.92
16	99.08	98.14	98.67	97.68	99.72	99.44	99.48	99.36
17	95.28	96.94	92.99	95.58	98.01	98.68	99.26	99.06
18	97.69	90.37	90.74	93.52	96.41	95.48	96.72	96.07
19	95.61	95.63	94.94	93.37	95.27	96.26	95.68	96.95
20	93.40	88.67	92.09	88.17	95.03	95.59	95.25	95.36
21	92.13	85.15	86.11	88.36	97.49	95.49	97.66	95.19
22	86.55	91.16	86.44	91.25	95.79	96.84	97.94	97.82
OA		96.24		96.73		98.29		98.52
AA	92.31	92.99	93.16	93.56	96.25	96.67	96.86	97.08
Kappa	96.24	96.18	96.73	96.72	98.29	98.28	98.52	98.51

results on the Pavia University dataset, Table VII shows the experimental results on the WHU-Hi-HongHu dataset, and Table VIII shows the experimental results on the WHU-Hi-Hanchuan dataset. In order to compare the performance between different methods in detail, we show the accuracy of each method in each category, as well as three indicators: overall accuracy, average accuracy, and Kappa coefficient. Comparing the experimental results on the three datasets, it can be found that the self-supervised pretraining model proposed by us has achieved very good results on each dataset.

Observing the experimental data on the PaviaU dataset, it can be found that traditional machine learning methods were the least effective, but SVM achieved over 90% accuracy in classification. The performance of deep learning was improved. The model obtained by pretraining with self-supervised learning gave the best classification results on the dataset. The difference between the correct classification results on the University of Pavia dataset was small, with close methods differing by around 2%. Among the deep learning methods, HybridSN's classification accuracy of 96.16% is still the best, while among the several self-supervised learning methods adopted, MAEST's accuracy is even lower than the average accuracy of only 91.77% of several supervised learning methods. The traditional masked self-supervised learning method HSIMAE achieves an accuracy of 97.36%, and the self-supervised contextual feature-based dual-branch Transformer method used in this chapter has further improved performance to 98.87%, which shows that the self-supervised pretraining with the addition of datasets has improved performance over the method using only separate datasets, and further validates the effectiveness of the improved structure.

WHU-Hi-HongHu dataset is a relatively difficult dataset, although the data volume of this dataset is sufficient, there are 22 kinds of landmark targets for classification, which is relatively difficult to distinguish. It can be found that none of the traditional machine learning methods on this dataset perform well, and the accuracy is below 90%, while the remaining deep learning methods all achieve accuracy rates above 90%. Among them, HybridSN achieved a final average accuracy of 96.76%, which is the method that achieved the best results besides the methods in this chapter. The traditional masked self-supervised learning method HSIMAE is only about 0.5% less accurate than HybridSN, while the method in this chapter still achieved the best classification results, and the performance was improved by more than 1% in comparison with the other best methods, and the average accuracy of classification reached 98.52%, which is an excellent classification result.

The experimental results on the WHU-Hi-Hanchuan dataset also further validate the previous experimental results. The method in this chapter also achieved superior results on the WHU-Hi-Hanchuan dataset, achieving the best performance for this classification, with an overall classification accuracy

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of 98.03%. However, there is not much difference between the classification performance on the Hanchuan dataset and the deep learning methods, where the best performing deep learning method is still HybridSN, which achieved an overall classification accuracy of 97.91%, with just less than 0.1% difference from the self-supervised method. We speculate that this is because the data in the dataset itself has good divisibility, which allows the deep learning method to get good performance as well, and the information obtained from the self-supervised pretraining can only serve as an aid to classification. While comparing self-supervised learning methods with each other, the method in this chapter still achieves the optimal classification results.

The above experimental results show that our proposed self-supervised learning method shows good performance on all datasets, and the multiple datasets used in the pretraining do not degrade the classification performance due to different data distribution, but can use various information in the pretraining dataset to improve the accuracy of the classification results. The accuracy improvement of this training method is obvious for small datasets, and it can also improve the classification accuracy by using supporting information when the data size increases.

In addition, to show and compare the classification result accuracy of the models more visually, we show the classification result maps of each model on the five datasets in Figs. 8–10. It can be found that the displayed results of the images are basically consistent with the experimental results, and the classification result maps generated by our method are basically consistent with the original labels. It should be noted that the clustering method of KNN can obtain a good classification map in the center of the region, but the effect in the edge and cross region is obviously not comparable to the deep learning method. Compared with the traditional methods, the neural network method can better distinguish the internal whole and edge information of each region in the image. Vision Transformer does not perform well as the CNN in terms of the overall image shape, but the classification result accuracy is relatively high and our method can provide smoother classification result maps on datasets with sufficient amount of data. The self-supervised pretraining model is able to obtain more complete classification maps with less misclassification noise, and the images are significantly better and smoother to look at.

In conclusion, our method has a relatively obvious advantage over current HSIC methods in that it is able to train and fine-tune hyperspectral classification models across datasets using large amounts of unlabeled hyperspectral image data, and the resulting models exhibit excellent performance on each dataset.

D. Ablation Experimental Results

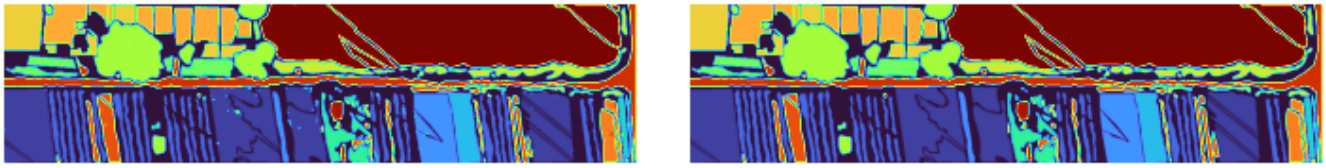
To better validate the performance of our method on multiple datasets, we controlled the number of participating pretraining datasets to analyze the effect of data volume on model performance, and also tested the effect of whether the dataset participated in pretraining on the subsequent fine-tuning performance. During this phase of the experiments, we mainly controlled whether the WHU-Hi-HanChuan dataset was involved in pretraining or not as the main variable. The complete self-supervised model was pretrained using all five datasets, and the ablation experimental model was pretrained using four datasets except WHU-Hi-HanChuan.

1) Data Volume Analysis: Due to the length of the article, we only show the classification results of two different models on the Indian Pines dataset and the WHU-Hi-Longkou dataset here, and that part of the classification results are shown in Tables IX and X. By analyzing the data in the two tables, it can be found that with pretraining using all five datasets, the classification results of the fine-tuned model on both datasets are better than using only four datasets. This performance improvement is more significant on the Indian Pines dataset, where the accuracy of the classification results on the Indian Pines dataset improves by about 1% when adding new data into the pretraining phase. On the WHU-Hi-Longkou dataset, the performance of the two classification models is almost similar, and they both perform well. It can be found that the classification results of the model are improved when more datasets are used for training, which further validates the utilization of data volume by our method to effectively extract more robust features from a large amount of data. The different performance on the two datasets further illustrates that our proposed self-supervised training method can achieve good classification results on small datasets by extending the pretraining dataset when facing smaller data volumes.

2) Pretraining Dataset Analysis: To further analyze the impact of whether the dataset is involved in pretraining on the model fine-tuning results, we show the classification performance of the two models on the WHU-Hi-Hanchuan dataset in Table XI. By comparing the data in the table, it is easy to see that the classification accuracy of the full model pretrained with the WHU-Hi-Hanchuan dataset is 0.5% higher than that of the model pretrained without the dataset. Due to the large size of the dataset, even a 0.5% improvement in classification performance is relatively significant for the classification of the data in the dataset. To show the difference in classification performance between the two models more intuitively, we show the classification result graphs of the two models on the dataset in Fig. 11. In the images, it can be found that the full model with the pretrained WHU-Hi-Hanchuan dataset has fewer classification spurious points and more complete and smooth image regions. Even when facing a large dataset like WHU-Hi-Hanchuan, adding the original data in the pretraining phase still results in better classification results. Since our method uses self-supervised learning in the pretraining phase, no labeled data is required, which makes it very easy to add a new dataset.

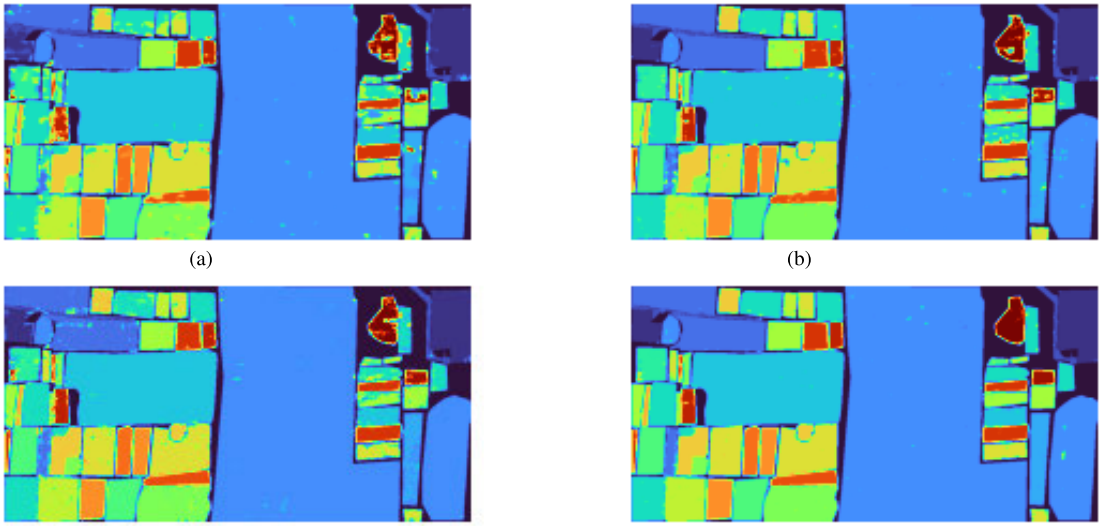
3) Methodological Validity Analysis: In order to better verify the effect of the method improvement in this article, a controlled improvement approach was taken in the ablation experiment stage for experimental comparison, using HSIMAE as the basic experimental baseline, the addition of two-branch Transformer and cross-attention mechanism and feature context-based self-supervision method were carried out, and the total number of control experiments was four,

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(a) (b)

Fig. 11. Model classification results comparison chart on the WHU-Hi-HanChuan dataset. The left figure shows the classification results of the pretrained classification model without the WHU-Hi-HanChuan dataset, while the right figure shows the classification results of the complete model, both of them have excellent classification performance. (a) Ablation experiment. (b) Complete model.



(a) (b) (c) (d)

Fig. 12. Model classification results comparison chart on the WHU-Hi-Honghu dataset. (a) HSIMAE + ViT. (b) HSIMAE + CATrans. (c) ContextMAE + ViT. (d) ContextMAE + CATrans.

which were baseline self-supervision method, addition of two-branch self-supervision method of backbone network, feature context self-supervision method using Vision Transformer, and HSIC method based on context self-supervision and null spectrum Transformer architecture proposed in this article. The datasets used in this experiment are all five datasets for pretraining and supervised fine-tuning on the downstream WHU-Hi-Honghu dataset as comparison results. The classification results of the four methods on the WHU-Hi-Honghu dataset are shown in Fig. 12.

In Table XII, the classification results of the four ablation experiments on the WHU-Hi-Honghu dataset are shown. Analyzing the results in the table, it can be found that the method performance of HSIMAE is excellent but still the lowest accuracy in the results of the ablation experiments, and the addition of the two-branch Transformer architecture to the method has a slight improvement in performance, and after changing the masked self-supervision to masked self-supervision of feature context, there is a significant improvement in the classification accuracy, with a 2% improvement in accuracy, at which time of the classification performance is already very close to that of the method in this chapter. The method proposed in this chapter still achieves the best classification results.

The experimental results show the two improvements in the self-supervised method in this article: the feature context-based self-supervised method and the two-branch spectral Transformer architecture have both improved the performance. Among them, the feature context-based self-supervision method improves significantly, the two-branch spectral Transformer improves but with relatively low effect, while the combination of the two methods can achieve the best results. This further proves the effectiveness of the proposed method in this article.

V. Conclusion

In this article, we have developed a method for self-supervised pretraining hyperspectral classification models across datasets. A channel adaptation module is proposed to solve the problem of band differences in hyperspectral cross datasets, and a masked self-supervised learning method is utilized to solve the label differences and labeling difficulties in training models across datasets. The proposed method can effectively utilize a large amount of data from different hyperspectral datasets to complete the training of classification models, and the fine-tuned training results on the downstream datasets have better classification performance compared with the current state-of-the-art classification models.

Our work shows that it is feasible to use hyperspectral data across datasets for model training, and large amount of data enables pretrained hyperspectral classification models to extract sufficiently robust features. One limitation is not to choose a backbone network that combines hyperspectral properties in order to verify the validity of the method. This part will be our future effort.

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